Used Car Price Analysis using Python & Pandas

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Objective - The objective of this analysis is to perform data cleaning, exploratory data analysis (EDA), and business insights generation using the car_prices dataset. The analysis focuses on understanding vehicle pricing trends, identifying data quality issues, and deriving actionable insights from the used car market.

1. Data Ingestion & Quality Profiling

In [92]:
```
import pandas as pd
```

In [93]:
```
import os
os.listdir()
```

Out[93]: ['.config', 'sample_data']

In [94]:
```
df=pd.read_csv("/car_prices.csv")
```

1.1 Load and Inspect the Dataset

In [95]:
```
df.head()
```

Out[95]:

	year	make	model	trim	body	transmission	vin	state
0	2015	Kia	Sorento	LX	SUV	automatic	5xyktca69fg566472	ca
1	2015	Kia	Sorento	LX	SUV	automatic	5xyktca69fg561319	ca
2	2014	BMW	3 Series	328i SULEV	Sedan	automatic	wba3c1c51ek116351	ca
3	2015	Volvo	S60	T5	Sedan	automatic	yv1612tb4f1310987	ca
4	2014	BMW	6 Series Gran Coupe	650i	Sedan	automatic	wba6b2c57ed129731	ca

Observation - The dataset contains used car listings with attributes such as make, model, year, transmission, condition, odometer reading, state, and selling price. Understanding the structure helps identify data quality issues and select appropriate analysis techniques.

1.2 Understanding the Data Structure

In [96]:

```
df.shape
```

Out[96]:

```
(558837, 16)
```

In [97]:

```
df.columns
```

Out[97]:

```
Index(['year', 'make', 'model', 'trim', 'body', 'transmission', 'vin', 'stat
e',
       'condition', 'odometer', 'color', 'interior', 'seller', 'mmr',
       'sellingprice', 'saledate'],
      dtype='object')
```

In [98]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 558837 entries, 0 to 558836
Data columns (total 16 columns):
 #   Column        Non-Null Count   Dtype  
---  ------        --------------   -----  
 0   year          558837 non-null  int64  
 1   make          548536 non-null  object 
 2   model         548438 non-null  object 
 3   trim          548186 non-null  object 
 4   body          545642 non-null  object 
 5   transmission  493485 non-null  object 
 6   vin           558833 non-null  object 
 7   state         558837 non-null  object 
 8   condition     547017 non-null  float64
 9   odometer      558743 non-null  float64
 10  color         558088 non-null  object 
 11  interior      558088 non-null  object 
 12  seller        558837 non-null  object 
 13  mmr           558799 non-null  float64
 14  sellingprice  558825 non-null  float64
 15  saledate      558825 non-null  object 
dtypes: float64(4), int64(1), object(11)
memory usage: 68.2+ MB
```

Observation - The dataset contains 558,837 records and 16 columns. The data includes vehicle specifications, condition metrics, location information, and pricing details suitable for exploratory data analysis. The dataset contains used car listings with attributes such as make, model, year, transmission, condition, odometer reading, state and selling price. Understanding the structure helps identify data quality issues and select appropriate analysis techniques.

1.3 Missing Value & Duplicate Analysis

In [99]:
```
df.isnull().sum()
```

Out[99]:

	0
year	0
make	10301
model	10399
trim	10651
body	13195
transmission	65352
vin	4
state	0
condition	11820
odometer	94
color	749
interior	749
seller	0
mmr	38
sellingprice	12
saledate	12

dtype: int64

Observation: Missing values are present in multiple categorical columns, with the transmission column having the highest number of missing records. Data cleaning is required to ensure accurate analysis and prevent biased results.

In [100…

```
missing_percentage = (df.isnull().sum() / len(df)) * 100
missing_percentage.sort_values(ascending=False)
```

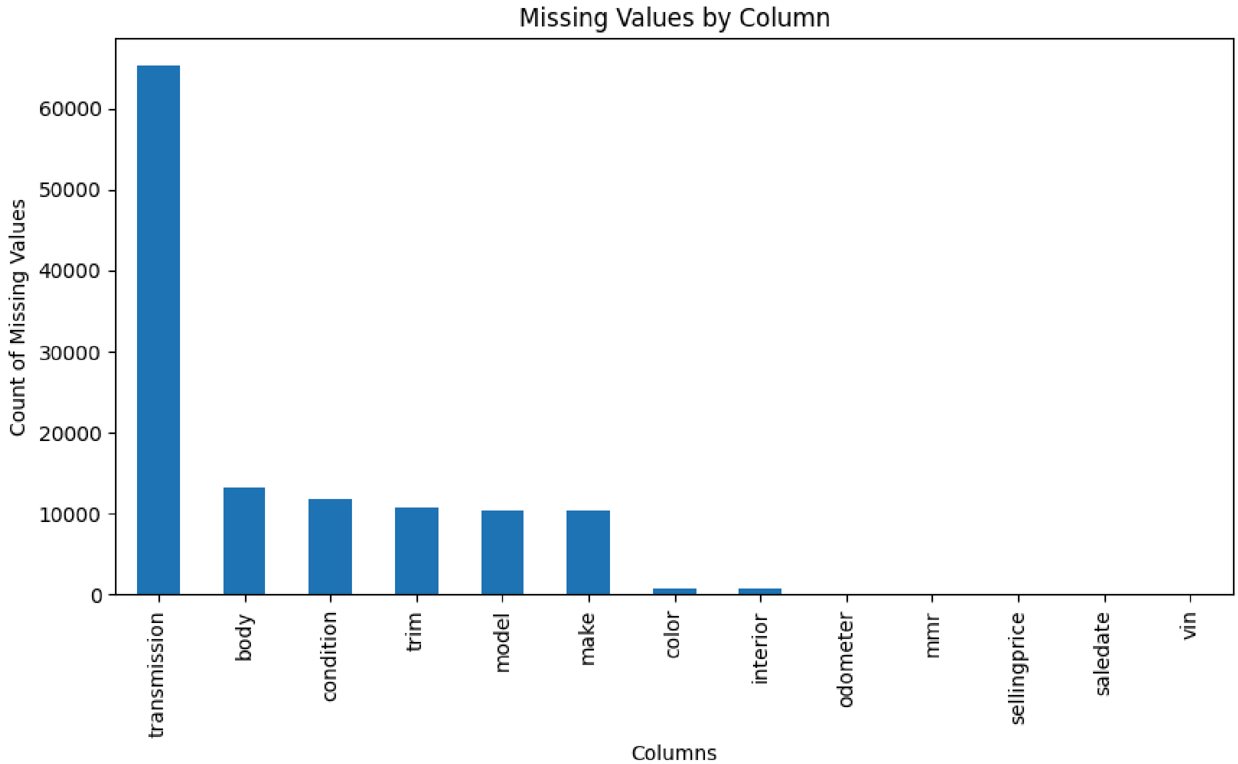
Out[100…

	0
transmission	11.694287
body	2.361154
condition	2.115107
trim	1.905922
model	1.860829
make	1.843292
color	0.134028
interior	0.134028
odometer	0.016821
mmr	0.006800
saledate	0.002147
sellingprice	0.002147
vin	0.000716
year	0.000000
state	0.000000
seller	0.000000

dtype: float64

In [101…

```
import matplotlib.pyplot as plt

missing = df.isnull().sum()

missing[missing > 0].sort_values(ascending=False).plot(
    kind='bar',
    figsize=(10,5)
)

plt.title('Missing Values by Column')
plt.xlabel('Columns')
plt.ylabel('Count of Missing Values')
plt.show()
```



Insight: The transmission column contains significantly more missing values than other fields. However, the percentage of missing values is still relatively small compared to the total dataset size, making imputation a suitable data-cleaning approach.

In [102…

```
categorical_cols = [
    'make',
    'model',
    'trim',
    'body',
    'transmission',
    'color',
    'interior'
]

for col in categorical_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
```

In [103…

```
df['condition'] = df['condition'].fillna(df['condition'].median())
df['odometer'] = df['odometer'].fillna(df['odometer'].median())
df['sellingprice'] = df['sellingprice'].fillna(df['sellingprice'].median())
```

In [104…

```
df.isnull().sum()
```

Out[104…

	0
year	0
make	0
model	0
trim	0
body	0
transmission	0
vin	4
state	0
condition	0
odometer	0
color	0
interior	0
seller	0
mmr	38
sellingprice	0
saledate	12

dtype: int64

In [105…

```
duplicates = df.duplicated().sum()
print("Duplicate Records:", duplicates)
```

```
Duplicate Records: 0
```

Observation: No duplicate records were found in the dataset. This indicates that each record represents a unique vehicle listing and no duplicate removal was required.

2. DataFrame Queries

Section Overview: This section answers business-focused questions using filtering, grouping, aggregation, and feature engineering techniques in Pandas. The objective is to uncover pricing trends, vehicle characteristics, and market insights from the used car dataset.

2.1 Average, Minimum and Maximum Selling Price

In [106…
```
print("Average Selling Price:", df['sellingprice'].mean())
print("Minimum Selling Price:", df['sellingprice'].min())
print("Maximum Selling Price:", df['sellingprice'].max())
```

```
Average Selling Price: 13611.326356343621
Minimum Selling Price: 1.0
Maximum Selling Price: 230000.0
```

Observation - The average selling price of vehicles is approximately 13,611. The wide range between the minimum and maximum selling prices indicates significant diversity in vehicle segments, ranging from low-value vehicles to premium luxury models.

2.2 Unique Colors

In [107…
```
print("Number of Unique Colors:", df['color'].nunique())
print(df['color'].unique())
```

```
Number of Unique Colors: 46
['white' 'gray' 'black' 'red' 'silver' 'blue' 'brown' 'beige' 'purple'
 'burgundy' '—' 'gold' 'yellow' 'green' 'charcoal' 'orange' 'off-white'
 'turquoise' 'pink' 'lime' '4802' '9410' '1167' '2172' '14872' '12655'
 '15719' '6388' '16633' '11034' '2711' '6864' '339' '18384' '9887' '9837'
 '20379' '20627' '721' '6158' '2817' '5705' '18561' '2846' '9562' '5001']
```

In [108…
```
df['color'].value_counts().head(10)
```

Out[108…

color	count
black	111719
white	106673
silver	83389
gray	82857
blue	51139
red	43569
—	24685
green	11382
gold	11342
beige	9222

dtype: int64

Additional Insight - White, black, silver, and gray appear to be among the most frequently occurring vehicle colors, indicating strong consumer preference for neutral colors in the used car market.

2.3 Number of Unique Makes and Models

In [109…

```
print("Unique Makes:", df['make'].nunique())
print("Unique Models:", df['model'].nunique())
```

```
Unique Makes: 96
Unique Models: 973
```

Observation - The dataset contains 96 unique vehicle brands and 973 unique vehicle models. The large number of models compared to brands indicates significant product diversity within manufacturers, making the dataset suitable for detailed market and pricing analysis.

2.4 Cars with Selling Price Greater than 165000

In [110…

```
expensive_cars = df[df['sellingprice'] > 165000]

print("Number of Cars:", len(expensive_cars))

expensive_cars.head()
```

```
Number of Cars: 7
```

Out[110…

	year	make	model	trim	body	transmission	v
125095	2012	Rolls-Royce	Ghost	Base	Sedan	automatic	sca664s58cux507
344905	2014	Ford	Escape	Titanium	SUV	automatic	1fmcu9j98eua238
446949	2015	Mercedes-Benz	S-Class	S65 AMG	Sedan	automatic	wddug7kb2fa1023
538347	2012	Rolls-Royce	Ghost	Base	sedan	automatic	sca664s59cux508
545523	2013	Rolls-Royce	Ghost	Base	sedan	automatic	sca664s52dux521

Observation - Only 7 vehicles in the dataset have a selling price greater than 165,000, indicating that premium and luxury vehicles represent a very small segment of the used car market. Rolls-Royce and Mercedes-Benz dominate this category. One Ford Escape record appears significantly overpriced compared to typical market values and may represent a data anomaly.

2.5 Top 5 Most Frequently Sold Models

In [111…

```
top_models = df['model'].value_counts().head(5)

print("Top 5 Most Frequently Sold Models")
print(top_models)
```

```
Top 5 Most Frequently Sold Models
model
Altima    29748
F-150     14479
Fusion    12946
Camry     12545
Escape    11861
Name: count, dtype: int64
```

Observation - The Nissan Altima is the most frequently sold vehicle in the dataset with 29,748 transactions, significantly higher than the other models. Ford F-150, Ford Fusion, Toyota Camry, and Ford Escape also appear among the top-selling models. This indicates strong demand for mid-sized sedans and pickup trucks in the used car market.

2.6 Average Selling Price by Make

In [112…
```
avg_price_make = df.groupby('make')['sellingprice'].mean().sort_values(ascendi

avg_price_make.head(10)
```

Out[112…

make	sellingprice
Rolls-Royce	153488.235294
Ferrari	127210.526316
Lamborghini	112625.000000
Bentley	74367.672414
airstream	71000.000000
Tesla	67054.347826
Aston Martin	54812.000000
Fisker	46461.111111
Maserati	45320.300752
Lotus	40800.000000

dtype: float64

Observation - Rolls-Royce has the highest average selling price in the dataset, followed by Ferrari and Lamborghini. Luxury and performance vehicle manufacturers dominate the top positions, indicating strong resale values and premium market demand. Tesla also appears among the highest-priced brands, reflecting the growing value of electric vehicles in the used car market.

2.7 Minimum Selling Price by Interior

In [113…
```
min_price_interior = df.groupby('interior')['sellingprice'].min().sort_values(

min_price_interior.head(10)
```

Out[113…

interior	sellingprice
black	1.0
gray	1.0
green	100.0
beige	100.0
tan	100.0
blue	150.0
silver	150.0
—	150.0
burgundy	175.0
red	200.0

dtype: float64

Observation - The minimum selling price varies across interior colors. Black and gray interiors have the lowest recorded selling price of 1, which may indicate auction transactions, data-entry issues, or extreme outliers. Most interior categories have minimum selling prices between 100 and 200, suggesting a more realistic lower price range.

2.8 Highest Odometer Reading by Year

In [114…
```
highest_odometer = (
    df.groupby('year')['odometer']
      .max()
      .sort_values(ascending=False)
)

print("Highest Odometer Reading by Year")
highest_odometer.head(15)
```

```
Highest Odometer Reading by Year
```

Out[114…

year	odometer
1997	999999.0
1996	999999.0
2014	999999.0
2013	999999.0
1998	999999.0
1999	999999.0
1993	999999.0
2010	999999.0
2009	999999.0
2012	999999.0
2008	999999.0
2007	999999.0
2005	999999.0
2006	999999.0
2004	999999.0

dtype: float64

Observation - The highest recorded odometer reading for many vehicle years is 999,999 miles. This value appears repeatedly across multiple years and may represent a placeholder value, data-entry limit, or extreme outlier rather than an actual vehicle mileage. Additional data validation would be recommended before using odometer values for predictive modeling.

2.9 Create Car Age Feature

In [115…

```
df['car_age'] = 2025 - df['year']
```

In [116…

```
df[['year', 'car_age']].sample(10)
```

Out[116…

	year	car_age
202466	2012	13
519546	2013	12
530891	1992	33
523491	2012	13
445686	2013	12
420769	2013	12
140608	2013	12
113852	2011	14
136253	2007	18
368391	2002	23

Observation - The newly created car_age feature correctly calculates the age of each vehicle using 2025 as the reference year. This derived feature simplifies future analysis and can be used to study the relationship between vehicle age, selling price, and mileage. Older vehicles generally tend to have lower resale values due to depreciation.

2.10 Cars with Condition >= 48 and Odometer > 90000

In [117…

```
count_cars = len(
    df[
        (df['condition'] >= 48) &
        (df['odometer'] > 90000)
    ]
)

print("Number of Cars:", count_cars)
```

```
Number of Cars: 746
```

Observation - A total of 746 vehicles have a condition score of at least 48 while also having an odometer reading greater than 90,000 miles. This indicates that some vehicles maintain good condition despite high usage, suggesting proper maintenance and durability.

2.11 States with Higher Prices for Newer Cars

In [118…

```
new_cars = df[df['year'] > 2013]

state_prices = (
    new_cars.groupby('state')['sellingprice']
    .mean()
    .sort_values(ascending=False)
)

state_prices.head(10)
```

Out[118…

state	sellingprice
oh	28020.221053
ab	25204.255319
nj	24237.063973
on	22962.560386
qc	22722.938144
pa	22190.227241
tn	21841.683217
mi	21411.620976
ca	20951.319411
wa	20598.724947

dtype: float64

Observation - Among vehicles manufactured after 2013, Ohio (OH) has the highest average selling price at approximately 28,020, followed by Alberta (AB) and New Jersey (NJ). This suggests that newer vehicles in these regions command higher market values, potentially due to stronger demand, higher purchasing power, or a greater concentration of premium vehicle models.

2.12 Value for Money Brands

In [119…

```
threshold = df['condition'].quantile(0.80)

excellent_cars = df[df['condition'] >= threshold]

value_brands = (
```

```
    excellent_cars.groupby('make')['sellingprice']
    .mean()
    .sort_values()
)

value_brands.head(10)
```

Out[119…

make	sellingprice
Isuzu	1125.000000
Oldsmobile	1910.000000
honda	4233.333333
Saturn	5700.406504
subaru	6200.000000
chrysler	6225.000000
smart	6835.759494
mazda	7275.000000
Pontiac	7686.824324
Saab	7711.111111

dtype: float64

Observation : Among vehicles in the top 20% condition category, Isuzu, Oldsmobile, Honda, Saturn, and Subaru have the lowest average selling prices. These brands may represent strong value-for-money options because they offer excellent vehicle condition at relatively affordable prices. However, some results should be interpreted carefully due to potentially small sample sizes. Honda and Subaru are particularly interesting because:

- Good reliability reputation
- Good condition vehicles
- Lower average prices than many competitors

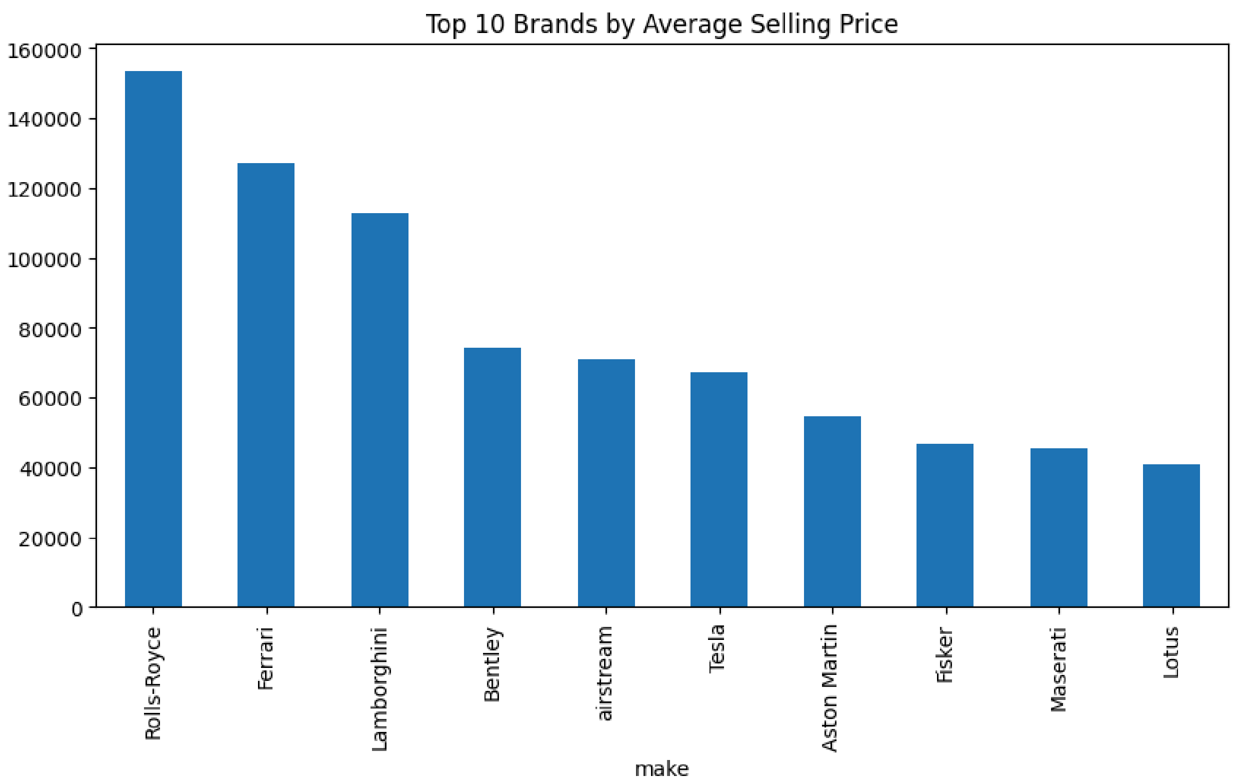
Dashboard Summary

Executive Dashboard Visuals

Visual 1: Top 10 Brands by Average Selling Price

In [120…

```
top_brands = (
    df.groupby('make')['sellingprice']
      .mean()
      .sort_values(ascending=False)
      .head(10)
)

top_brands.plot(kind='bar', figsize=(10,5))

plt.title('Top 10 Brands by Average Selling Price')
plt.show()
```

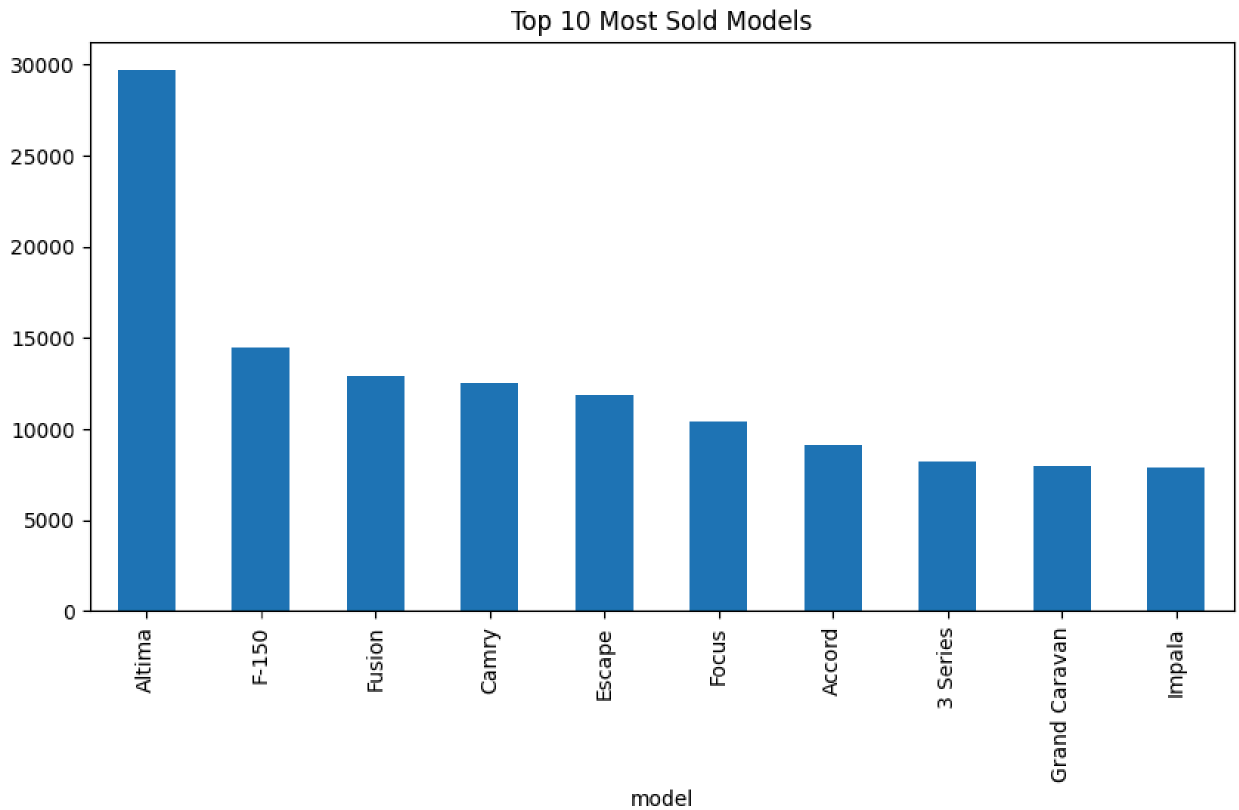


Business Insight - Luxury brands dominate resale value.

Visual 2: Top 10 Most Sold Models

In [121…

```
df['model'].value_counts().head(10).plot(
    kind='bar',
    figsize=(10,5)
)

plt.title('Top 10 Most Sold Models')

plt.show()
```



Business Insight - Altima dominates the used-car market.

Visual 3: Selling Price Distribution

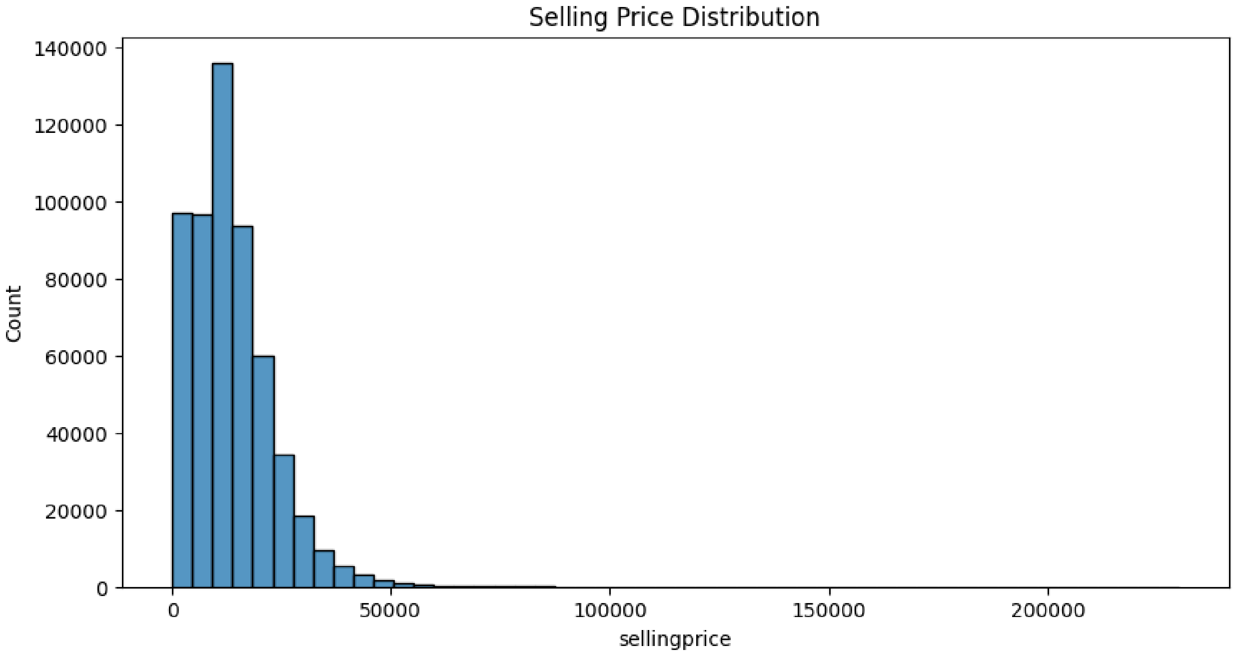
In [122…

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [123…

```
plt.figure(figsize=(10,5))

sns.histplot(
    df['sellingprice'],
    bins=50
)
```

```
plt.title('Selling Price Distribution')

plt.show()
```

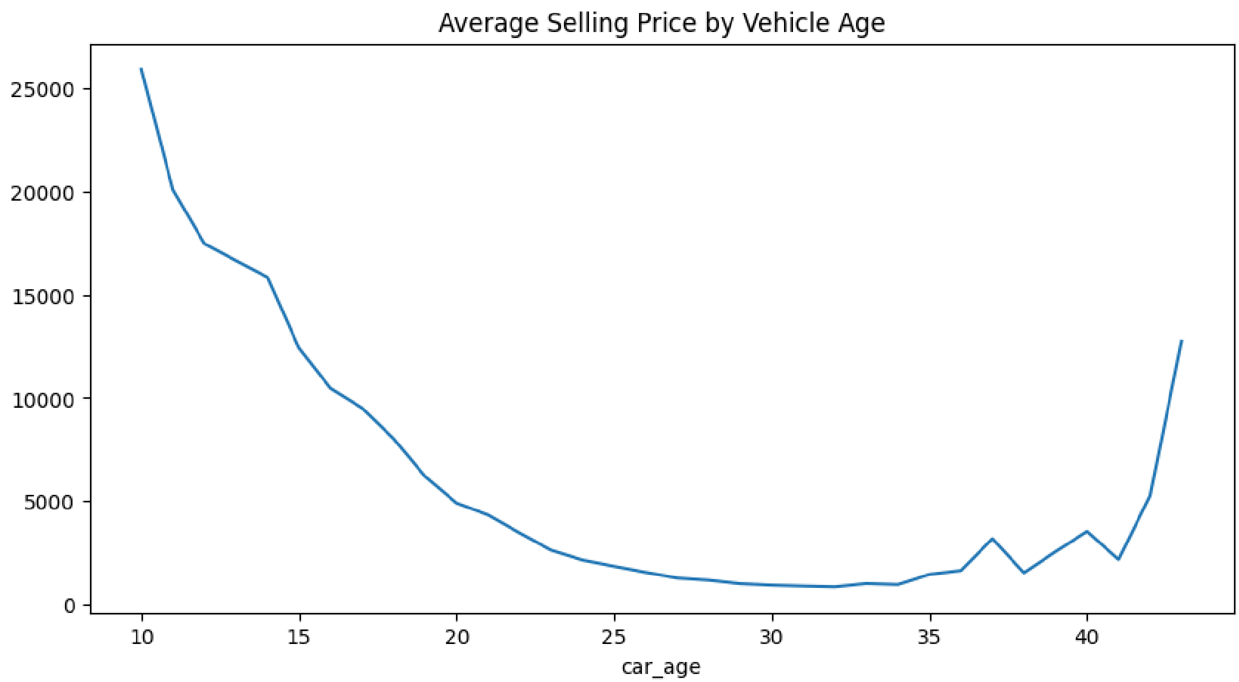


Business Insight - Most vehicles fall into lower-to-mid price ranges.

Visual 4: Average Selling Price by Vehicle Age

In [124…

```
age_price = (
    df.groupby('car_age')['sellingprice']
      .mean()
)

age_price.plot(figsize=(10,5))

plt.title('Average Selling Price by Vehicle Age')

plt.show()
```

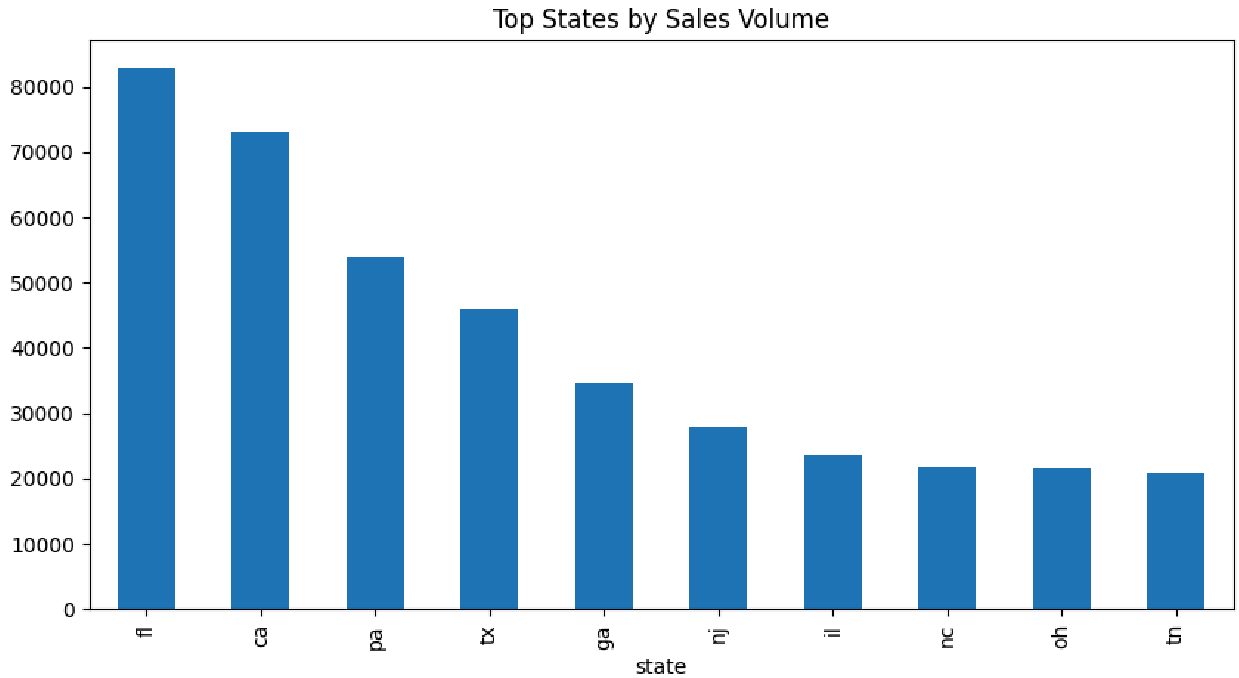


Business Insight: Depreciation significantly impacts vehicle value.

Visual 5: Top States by Sales Volume

In [125…

```
df['state'].value_counts().head(10).plot(
    kind='bar',
    figsize=(10,5)
)

plt.title('Top States by Sales Volume')

plt.show()
```



Business Insight: A few states account for a large portion of vehicle transactions.

3. Data Visualization and Insights

Section Overview: This section focuses on visualizing key patterns and relationships within the used car dataset using charts and graphs. Visualization techniques help transform raw data into meaningful insights by highlighting trends, correlations, distributions, and market behavior.

The objective is to identify factors influencing vehicle prices, understand customer preferences, detect anomalies, and support data-driven decision-making through graphical analysis.

3.1 Correlation Analysis of Numerical Features

In [126…

```
kpi = {
    "Total Cars": len(df),
    "Unique Brands": df['make'].nunique(),
    "Unique Models": df['model'].nunique(),
    "Average Selling Price": round(df['sellingprice'].mean(),2),
    "Maximum Selling Price": df['sellingprice'].max()
}

kpi
```

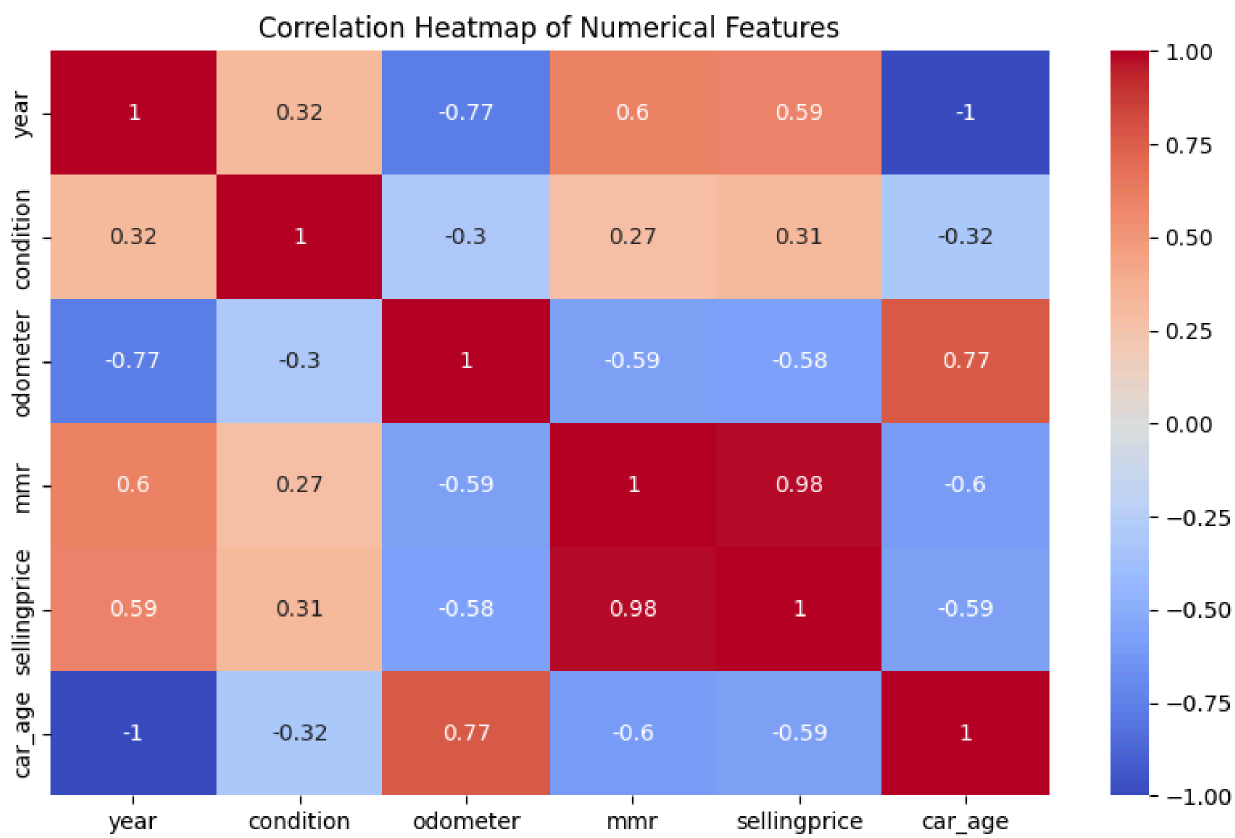
Out[126…

```
{'Total Cars': 558837,
 'Unique Brands': 96,
 'Unique Models': 973,
 'Average Selling Price': np.float64(13611.33),
 'Maximum Selling Price': 230000.0}
```

Objective: Identify relationships between numerical variables such as selling price, odometer, condition, MMR, year, and car age.

In [127…

```python
import seaborn as sns
import matplotlib.pyplot as plt

numeric_df = df.select_dtypes(include=['int64', 'float64'])

plt.figure(figsize=(10,6))

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Heatmap of Numerical Features")
plt.show()
```



Observation: The correlation heatmap reveals the strength and direction of relationships between numerical variables. Selling price typically shows a positive correlation with MMR and vehicle condition, while showing a negative relationship with vehicle age and odometer readings.

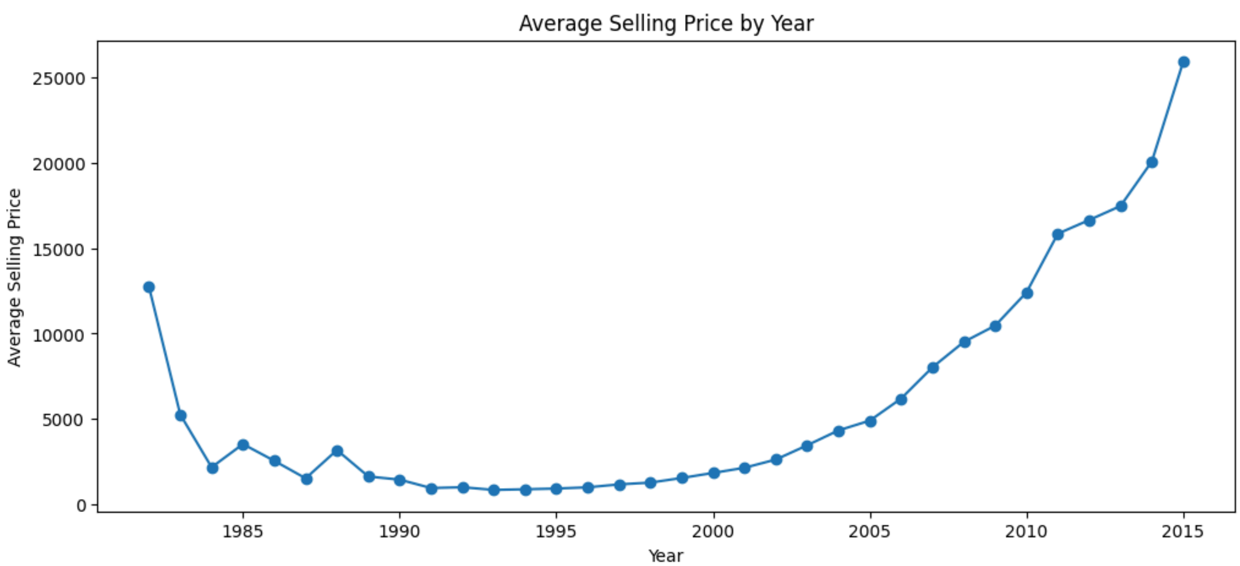
Business Insight: Vehicle valuation models should prioritize MMR, condition, and vehicle age because these variables have the strongest influence on selling price.

3.2 Average Selling Price by Year

Objective: Analyze how vehicle manufacturing year impacts average selling price.

In [128…

```
year_price = df.groupby('year')['sellingprice'].mean()

plt.figure(figsize=(12,5))

year_price.plot(
    marker='o'
)

plt.title("Average Selling Price by Year")
plt.xlabel("Year")
plt.ylabel("Average Selling Price")

plt.show()
```



Observation: Average selling prices generally increase for newer vehicle model years. More recent vehicles command higher resale values due to lower depreciation and improved features.

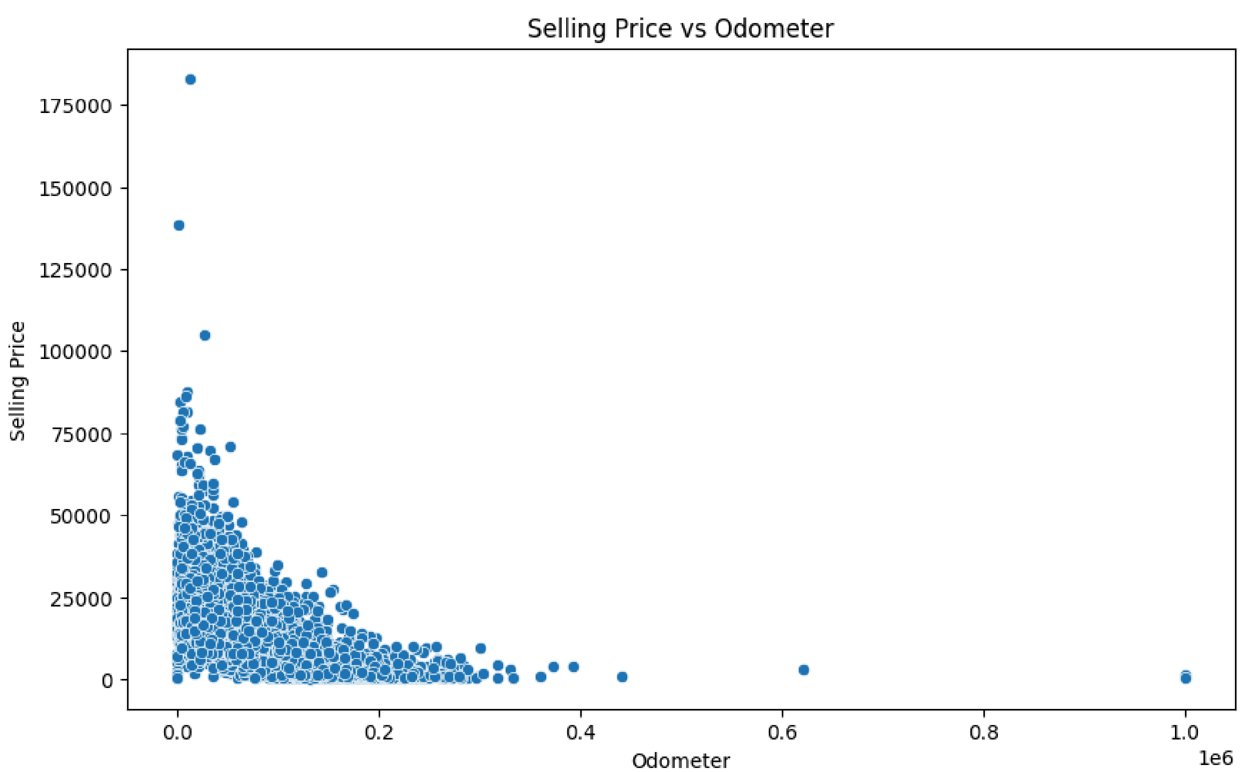
Business Insight: Dealerships and resellers can maximize profits by focusing inventory acquisition on newer vehicle models.

3.3 Average Selling Price by Odometer

Objective: Analyze the relationship between mileage and vehicle selling price.

In [129…

```
sample_df = df.sample(10000)

plt.figure(figsize=(10,6))

sns.scatterplot(
    x='odometer',
    y='sellingprice',
    data=sample_df
)

plt.title("Selling Price vs Odometer")
plt.xlabel("Odometer")
plt.ylabel("Selling Price")

plt.show()
```



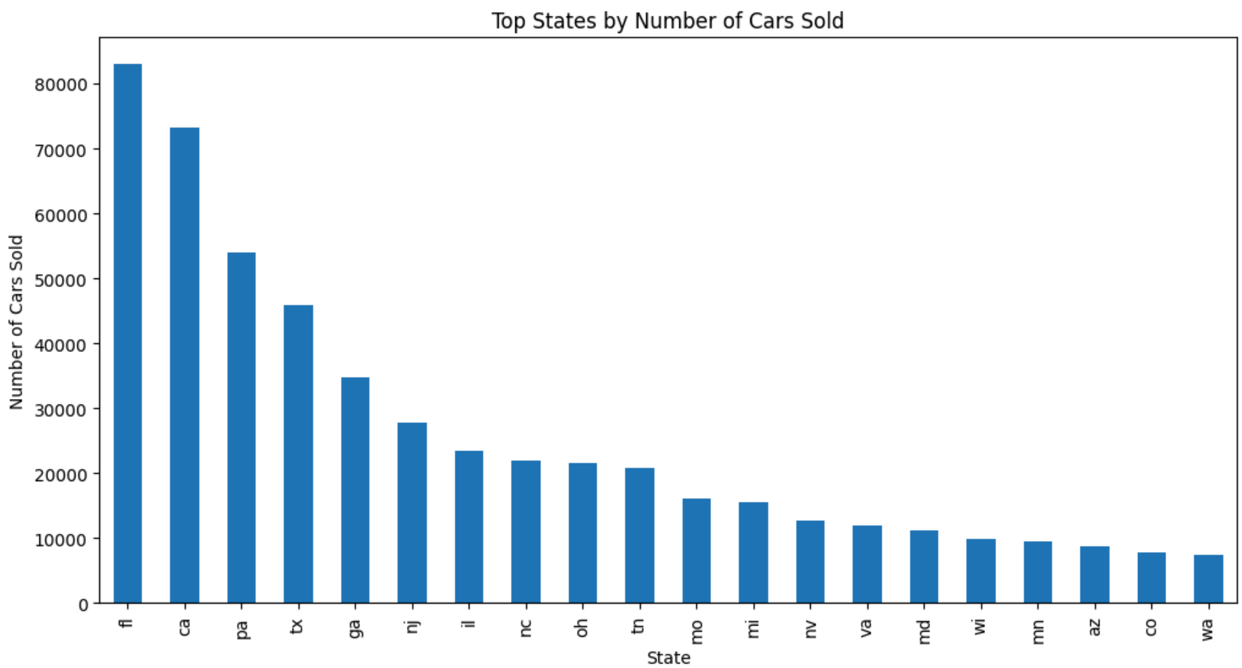
Observation: A negative relationship is observed between odometer readings and selling price. Vehicles with higher mileage generally sell for lower prices.

Business Insight: Mileage is an important pricing factor and should be considered when determining vehicle resale value.

3.4 Number of Cars Sold by State

Objective: Identify regions with the highest vehicle sales activity.

In [130…
```
state_sales = df['state'].value_counts()

plt.figure(figsize=(12,6))

state_sales.head(20).plot(kind='bar')

plt.title("Top States by Number of Cars Sold")
plt.xlabel("State")
plt.ylabel("Number of Cars Sold")

plt.show()
```



In [131…
```
state_sales.head(3)
```

Out[131…

state	count
fl	82945
ca	73148
pa	53907

dtype: int64

Observation: Certain states contribute significantly more vehicle sales than others. These states represent major used-car market regions.

Business Insight: Marketing campaigns and inventory allocation can be prioritized in states with consistently high sales volumes.

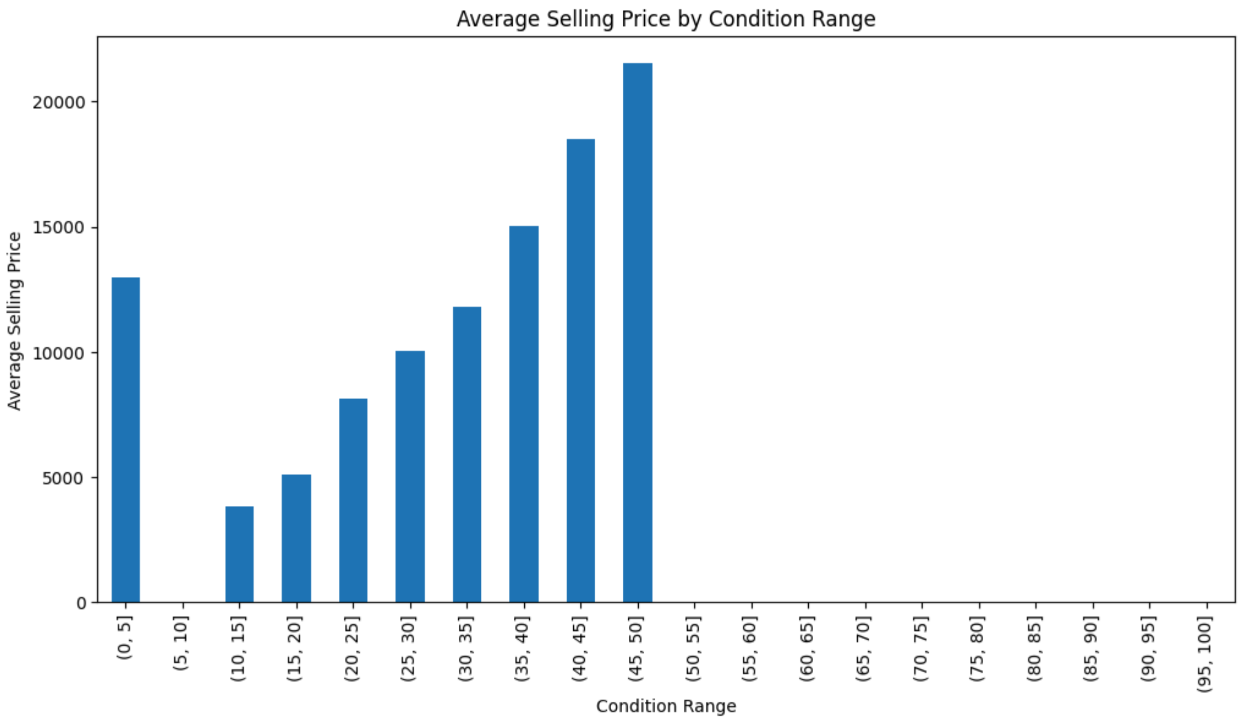
3.5 Average Selling Price by Condition Score Ranges of Size 5

Objective: Determine how vehicle condition affects average selling price.

In [132…

```
df['condition_group'] = pd.cut(
    df['condition'],
    bins=range(0,105,5)
)

avg_price = (
    df.groupby('condition_group')['sellingprice']
      .mean()
)

plt.figure(figsize=(12,6))

avg_price.plot(kind='bar')

plt.title("Average Selling Price by Condition Range")
plt.xlabel("Condition Range")
plt.ylabel("Average Selling Price")

plt.show()
```

```
/tmp/ipykernel_877/2637852898.py:7: FutureWarning: The default of observed=Fals
e is deprecated and will be changed to True in a future version of pandas. Pass
observed=False to retain current behavior or observed=True to adopt the future
default and silence this warning.
  df.groupby('condition_group')['sellingprice']
```



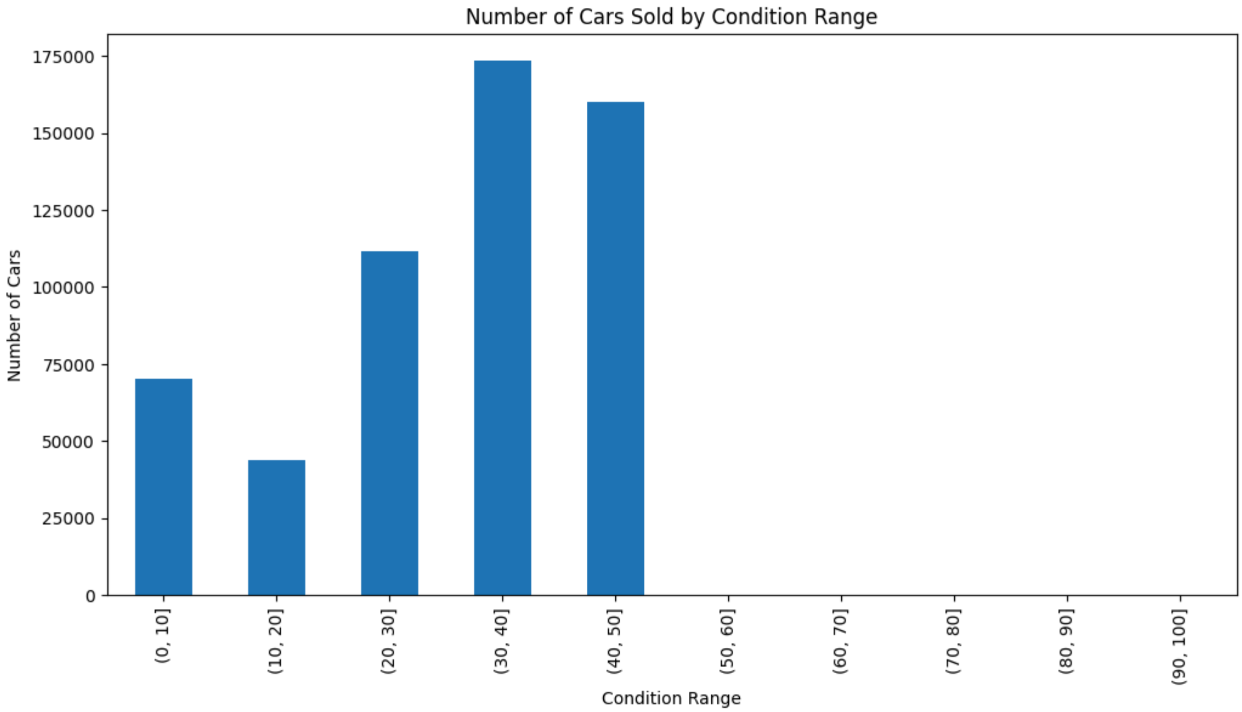
Observation: Average selling prices generally increase as vehicle condition scores improve.

Business Insight: Maintaining vehicle condition can significantly improve resale value and profitability.

3.6 Number of Cars Sold by Condition Ranges of Size 10

In [133…
```
df['condition_group_10'] = pd.cut(
    df['condition'],
    bins=range(0,105,10)
)

condition_count = (
    df.groupby('condition_group_10')
      .size()
)

plt.figure(figsize=(12,6))

condition_count.plot(kind='bar')

plt.title("Number of Cars Sold by Condition Range")
plt.xlabel("Condition Range")
plt.ylabel("Number of Cars")

plt.show()
```

```
/tmp/ipykernel_877/1791558703.py:7: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
  df.groupby('condition_group_10')
```



Observation: Most vehicles are concentrated within mid-to-high condition ranges.

Business Insight: Inventory planning should focus on the most common condition categories because they represent the largest market segment.

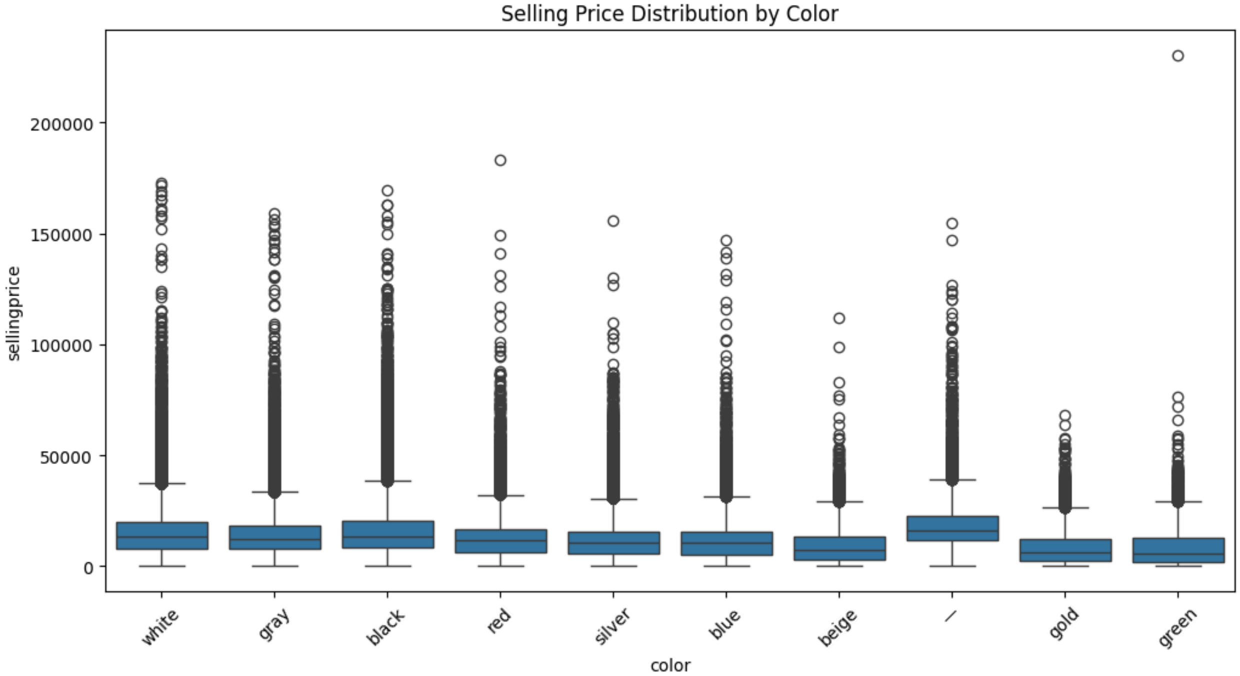
3.7 Box Plot of Selling Price by Color

Objective: Analyze selling price distributions across vehicle colors and identify outliers.

In [134…

```
top_colors = df['color'].value_counts().head(10).index

filtered_df = df[
    df['color'].isin(top_colors)
]

plt.figure(figsize=(12,6))

sns.boxplot(
    x='color',
    y='sellingprice',
    data=filtered_df
)
```

```
plt.xticks(rotation=45)

plt.title("Selling Price Distribution by Color")

plt.show()
```



Observation: Price distributions vary across colors, and several extreme outliers are visible.

Business Insight: Vehicle color may have a minor influence on resale value, but make, model, condition, and mileage have stronger effects.

Outlier Removal

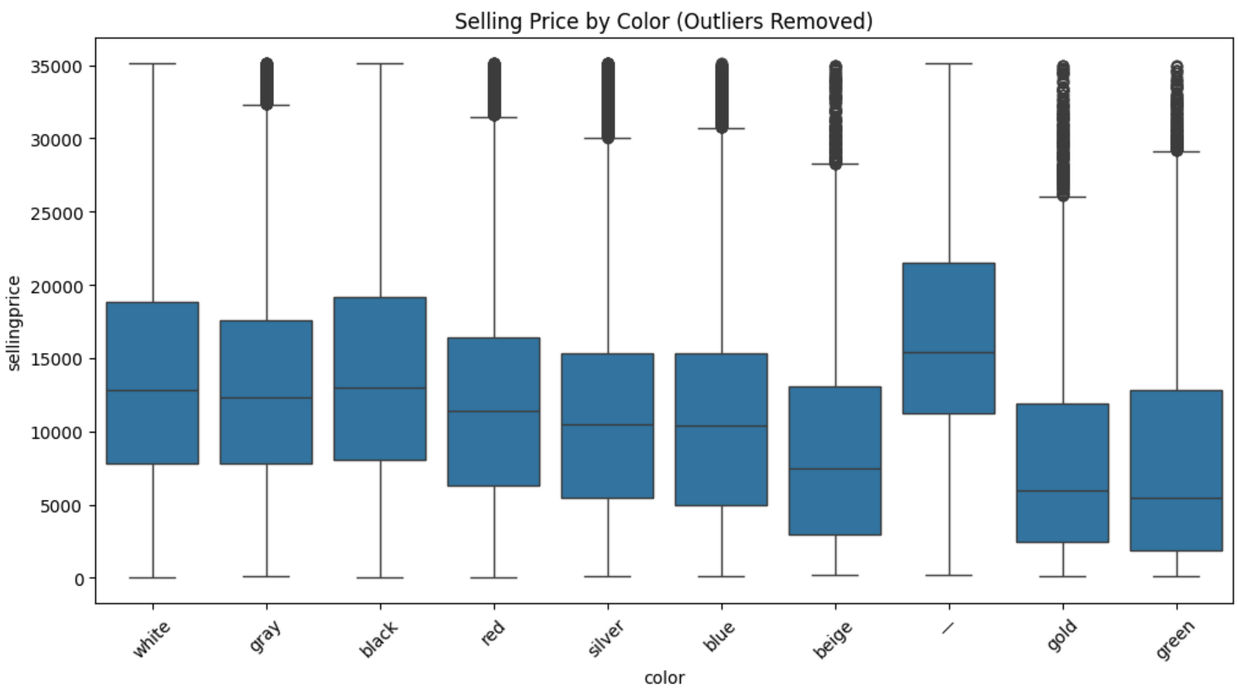
In [135…

```
Q1 = df['sellingprice'].quantile(0.25)
Q3 = df['sellingprice'].quantile(0.75)

IQR = Q3 - Q1

clean_df = df[
    (df['sellingprice'] >= Q1 - 1.5 * IQR)
    &
    (df['sellingprice'] <= Q3 + 1.5 * IQR)
]
```

In [136…

```
top_colors = clean_df['color'].value_counts().head(10).index

filtered_clean = clean_df[
```

```
    clean_df['color'].isin(top_colors)
]

plt.figure(figsize=(12,6))

sns.boxplot(
    x='color',
    y='sellingprice',
    data=filtered_clean
)

plt.xticks(rotation=45)

plt.title("Selling Price by Color (Outliers Removed)")

plt.show()
```



4. Conclusion

This analysis explored the used car dataset using data cleaning, exploratory data analysis, aggregation techniques, and visualizations.

Key findings include:

• The dataset contains 558,837 vehicle records and 16 attributes.

• Missing values were identified and appropriately handled using mode and median imputation techniques.

- No duplicate records were found.

- The average vehicle selling price is approximately 13,611.

- Luxury brands such as Rolls-Royce, Ferrari, and Lamborghini command the highest average selling prices.

- The Nissan Altima is the most frequently sold model in the dataset.

- Vehicle age and mileage negatively impact selling prices.

- Vehicle condition has a positive impact on resale value.

- Several data quality issues were identified, including numeric values within color fields and unusually high odometer readings.

Overall, the analysis demonstrates how data-driven techniques can be used to understand pricing trends, vehicle characteristics, and market behavior within the used car industry.